dimension in, say, the type IIA formulation, one-dimensional strings stretch into two-dimensional inner-tube-like membranes. If you assume the tenth dimension is very small, as had always been implicitly done prior to these discoveries, the inner tubes look and behave like strings. As is the case for strings, the question of whether these newly found branes are indivisible or, instead, are made of yet finer constituents, remains unanswered. Researchers are open to the possibility that the ingredients so far identified in string/M-theory will not bring to a close the search for *the* elementary constituents of the universe. However, it's also possible that they will. Since much of what follows is insensitive to this issue, we'll adopt the simplest perspective and imagine that all the ingredients—strings and branes of various dimensions—are fundamental. And what of the earlier reasoning, which suggested that fundamental higher dimensional objects could not be incorporated into a physically sensible framework? Well, that reasoning was itself rooted in another quantum mechanical approximation scheme—one that is standard and fully battle tested but that, like any approximation, has limitations. Although researchers have yet to figure out all the subtleties associated with incorporating higher-dimensional objects into a quantum theory, these ingredients fit so perfectly and consistently within all five string formulations that almost everyone believes that the feared violations of basic and sacred physical principles are absent.

4. In fact, we could be living on an even higher-dimensional brane (a four-brane, a five-brane . . .) three of whose dimensions fill ordinary space, and whose other dimensions fill some of the smaller, extra dimensions the theory requires.

5. The mathematically inclined reader should note that for many years string theorists have known that closed strings respect something called T-duality (as explained further in Chapter 16, and in Chapter 10 of *The Elegant Universe*). Basically, T-duality is the statement that if an extra dimension should be in the shape of a circle, string theory is completely insensitive to whether the circle's radius is R or $1/R$. The reason is that strings can move around the circle ("momentum modes") and/or wrap around the circle ("winding modes") and, under the replacement of R with $1/R$, physicists have realized that the roles of these two modes simply interchange, keeping the overall physical properties of the theory unchanged. Essential to this reasoning is that the strings are closed loops,